Suggested problems: 1-13 odd

Problem 1: The table below contains partial values for a linear function. Fill in the blanks.

$x \backslash y$	-1.0	0.0	1.0
2.0	4.0		
3.0		3.0	5.0

First row, left to right: 6, 8. Second row: 1.

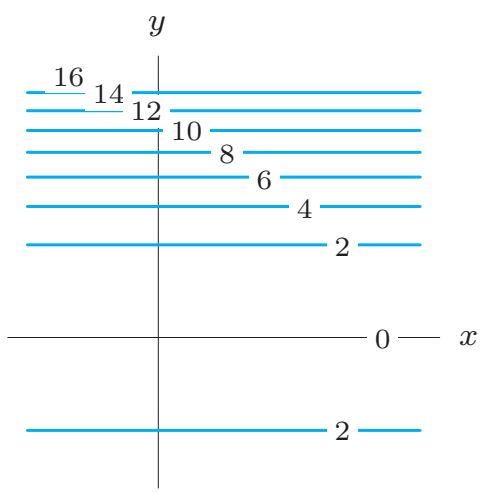
Problem 2: Find an equation of the plane passing through the points $(1, 0, 1)$, $(1, -1, 3)$, and $(3, 0, -1)$.

An equation is

$$z - 1 = -(x - 1) - 2(y - 0).$$

See page 695 in the online textbook, and class notes.

Problem 3: Could the diagram below represent the contour diagram of a linear function?



No. For example, moving upward from the 0 contour line, each contour line passed represents an increase in height of 2 units. But the contour lines are getting closer together, which means the graph is getting steeper. Thus the slope in the y-direction is not constant.